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Palaeomagnetic studies of Proterozoic rocks from the Lake Onega region, southeast Fennoscandian Shield

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SUMMARY

In a recent compilation, Elming *et al.* (1993) presented a new apparent polar wander path for the Proterozoic of the Fennoscandian Shield. However, only a minor portion of the data used for this and for the earlier compilations has been obtained from the eastern part of the Fennoscandian Shield, thus prompting the present study.

In this paper we present new palaeomagnetic data for the Lake Onega region. The sections studied comprise quartzites, as well as extrusive and intrusive rocks of Vepsian (1900–1650 Ma) and Jatulian (2300–2100 Ma) ages, which represent Svecofennian and Jatulian tectonomagmatic events respectively. Thermal and alternating-field demagnetization experiments reveal the presence of several components of the natural remanent magnetization of normal and reversed polarity. Component A_N has been identified in a sill of Vepsian gabbro-dolerites (1770 Ma) exposed at Rybreka locality and yields a palaeopole at 37.8°N, 210.6°E (dp = 4.0°, dm = 7.7°). This magnetization is interpreted to be of thermoremanent origin. In the Vepsian Shoksha quartzite (*ca.* 1800 Ma) cut by the sill and in the underlying basalts, the palaeopoles indicated by the presumably primary components A_N and A_R are at 45.1°N, 206.4°E (dp = 3.6°, dm = 6.5°) and 44.2°N, 241.5°E (dp = 6.1°, dm = 9.6°) respectively. Component A_N was isolated in the Jatulian basaltic lavas (*ca* 2200–2300 Ma) at Girvas locality also; however in this case it appears as secondary and yields nearly the same palaeopole: 43.1°N, 211.5°E (dp = 4.9°, dm = 8.8°). Component B_R was isolated in Vepsian gabbro-dolerites and quartzites, sampled from zones altered by monzonite veins representing the final phase of sill emplacement. This component yields a palaeopole at 13.9°N, 171.3°E (dp = 2.4°, dm = 4.8°) for gabbro-dolerites and at 10.5°N, 201.8°E (dp = 8.4°, dm = 14.9°) for quartzites. The Vepsian and Jatulian rocks also show the presence of presumably younger (< 1500 Ma) magnetization components *C*, *D* and *E*. The presumably primary stable dual-polarity components determined in the Jatulian lavas and interlayers of the red siliceous rocks yield a palaeopole at 0.8°S, 245.4°E (dp = 7.2°, dm = 11.6°) for the upper part of the section (component *F*) and a palaeopole at 36.0°S, 307.4°E (dp = 9.9°, dm = 14.9°) for the lower part of the section (component *G*). These palaeomagnetic poles are compared with the apparent polar wander path of Fennoscandia for the interval 2400–1200 Ma (after Elming *et al.* 1993) and with some recent results from Vepsian and Jatulian rocks.

Key words: Early Proterozoic, Fennoscandia, palaeomagnetism, pole positions, south Karelia.

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1 INTRODUCTION

The eastern part of the Fennoscandian Shield has been a target for palaeomagnetic studies since the early work of Katzeblin (1968). At that time, the palaeomagnetism of the Proterozoic rocks in Finland and Sweden was also examined (Neuvonen 1965; Cornwell 1968; Pesonen & Stigzelius 1972). After these early works, the Precambrian formations of this western part of the shield were studied extensively (e.g. Piper 1979; Bylund & Pesonen 1987), forming the main part of the database for construction of the Proterozoic apparent polar wander path (APWP) for the Fennoscandian Shield (Pesonen & Neuvonen 1981; Bylund 1985; Pesonen *et al.* 1989; Mertanen & Pesonen 1990; Pesonen *et al.* 1991; Elming *et al.* 1993). As concerns the eastern part of the shield, the situation appears to be quite different: only 14 palaeomagnetic poles for the Precambrian formations of the Karelian–Kola region have been included in the recent Fennoscandian palaeomagnetic database, in comparison with 220 poles for the western part of the shield (Pesonen *et al.* 1991). Although there are some new data (Krasnova & Gooskova 1995; Torsvik & Meert 1995; Mertanen 1995), this does not change the situation fundamentally. In order to improve this situation and to assess the reliability of the earlier data, a cooperative palaeomagnetic study has been undertaken.

Sequences of Lower Proterozoic rocks in the Lake Onega region (both sedimentary and igneous rocks) covering the Vepsian (1850–1650 Ma) and Jatulian (2300–2150 Ma) ages, originally studied by Katzeblin (1968), Ignatjeva (1974), Khramov & Ignatjeva (1980) and Gooskova & Krasnova (1985), have been resampled.

2 GEOLOGICAL SETTING AND SAMPLING

The sampling region lies close to the west and northwest shore line of Lake Onega, in the southeast Fennoscandian Shield (Fig. 1). This region belongs to the Onega depression, which lies on the southeast side of the Karelian tectonic province (Sokolov 1987).

The Vepsian (1850–1650 Ma) age sequence comprises two members. The lowest is the Petrozavodskian Sandstone Formation which has yielded a K–Ar age (on glauconite) of 1800 Ma (Sokolov 1987). Two basaltic lava flows occur in the upper part of this member. These rocks are overlain by the Shoksha Quartzite Formation, which has been intruded by a

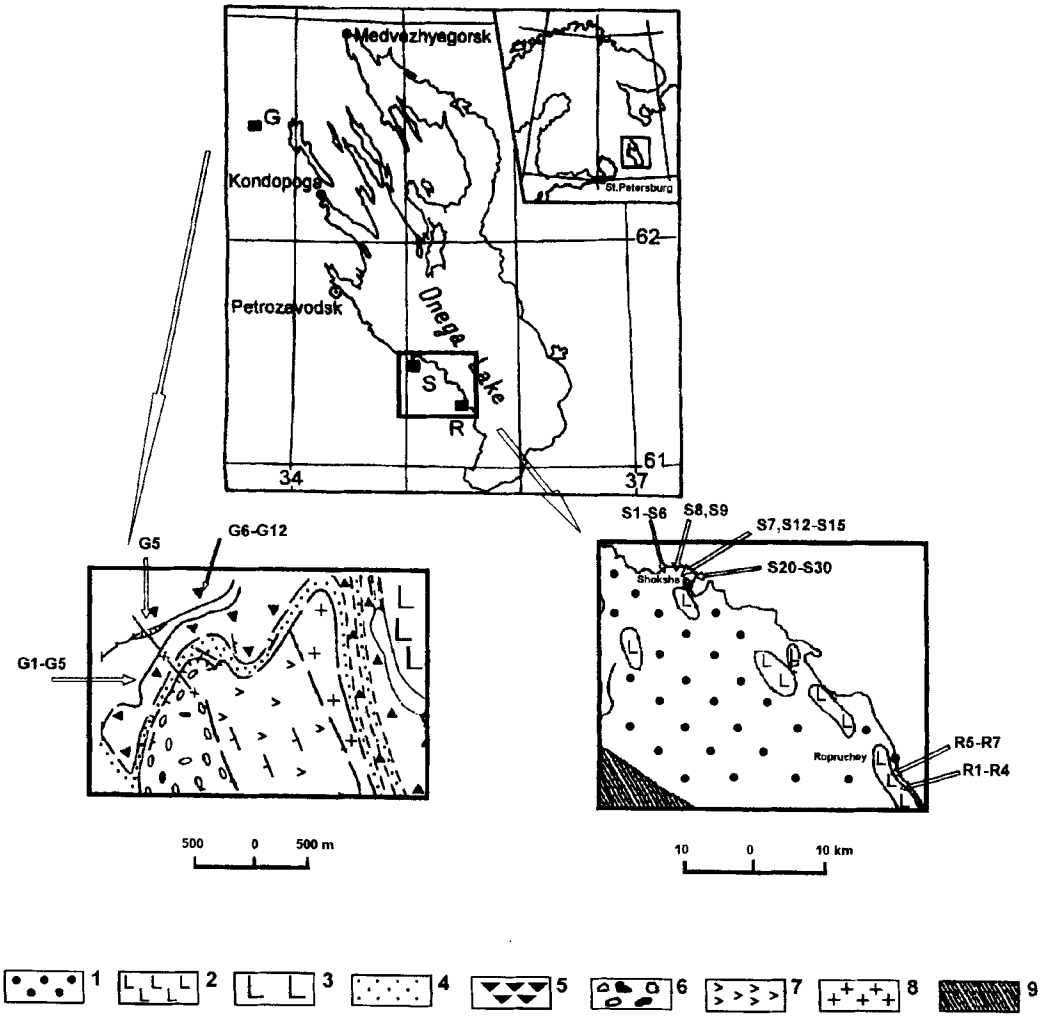


Figure 1. Map of sampling localities. G-Girvas, S-Shoksha, R-Rybreka. Vepsian: 1: quartzites and sandstones; 2: gabbro-dolerites. Jatulian: 3: gabbro; 4: quartzites; 5: basalts. Sariolian: 6: conglomerates; 7: andesitic basalts. Archaean: 8: granites; 9: Palaeozoic sedimentary cover.

gabbro-dolerite sill of Vepsian age (1770 ± 12 Ma. U–Pb on zircon: Bibikova *et al.* 1990). This sill is part of a vast layered intrusion, 180 m in thickness and 150 km in lateral extent. For this study, samples were collected from the layered intrusion near the village of Rybreka (sites R1 to R7, Figs 1 and 2), from the Shoksha Quartzite Formation quarried to the north of the village of Shoksha (sites S1, S2, S4, S6, S8, S9, S12-15, S20-30) and from a basaltic lava flow (sites S7 and 7a) near this quarry (Figs 1 and 2).

Undeformed subhorizontal (tilted to the north at 12°–17°) Jatulian-age lava flows are exposed to the north of Lake Onega (Girvas sites). Their ages are constrained by two U–Pb isochrons: 2420 ± 45 Ma for pebbles from the basal conglomerates and 2150 ± 60 Ma for the cross-cutting gabbro-dolerite dykes (Sokolov 1987). 29 samples were collected from the exposures along the Suna River east of the village of Girvas (sites G1, G1a, G2, G3, G4, G7, G9, G10, G11, G12, Fig. 2). In addition, interbedded haematized siliceous rocks were also sampled (seven samples from sites G5, G8). The lava flows are all of Middle Jatulian age except the lowermost 40-m thick andesitic basalt flow, which is Lower Jatulian (Fig. 2).

During the field work, two to four block samples were taken from each site. Magnetic and sun compasses were used for core orientation.

3 LABORATORY PROCEDURE

After joint field work, the block samples were drilled (one 10–15 cm core from each sample). Core samples were then distributed among the participating laboratories for standard palaeomagnetic measurements. A pilot collection (18 samples), representative of all rock types sampled, was subjected to a variety of rock-magnetic experiments at the Institute of Physics of the Earth in Moscow. These experiments include determinations of saturation magnetization J_s and saturation remanence J_{rs} as a function of temperature, as well as AF and DC demagnetization of J_{rs}. The peak fields used for the latter two experiments reach 100 mT and 2000 mT respectively. In addition, stepwise thermal and AF demagnetization experiments were carried out on specimens from the pilot collection. A magnetically shielded furnace (inner residual field < 15 nT) was used for the demagnetization experiments, whereas AF demagnetization was achieved by using a two-axis tumbler system, housed in a set of Helmholtz coils in order to compensate for the Earth's local magnetic field in the laboratory. In the case of stronger remanence, continuous thermal demagnetization was carried out in a thermal demagnetizer incorporated in the spinner magnetometer.

Routine palaeomagnetic measurements of the bulk of

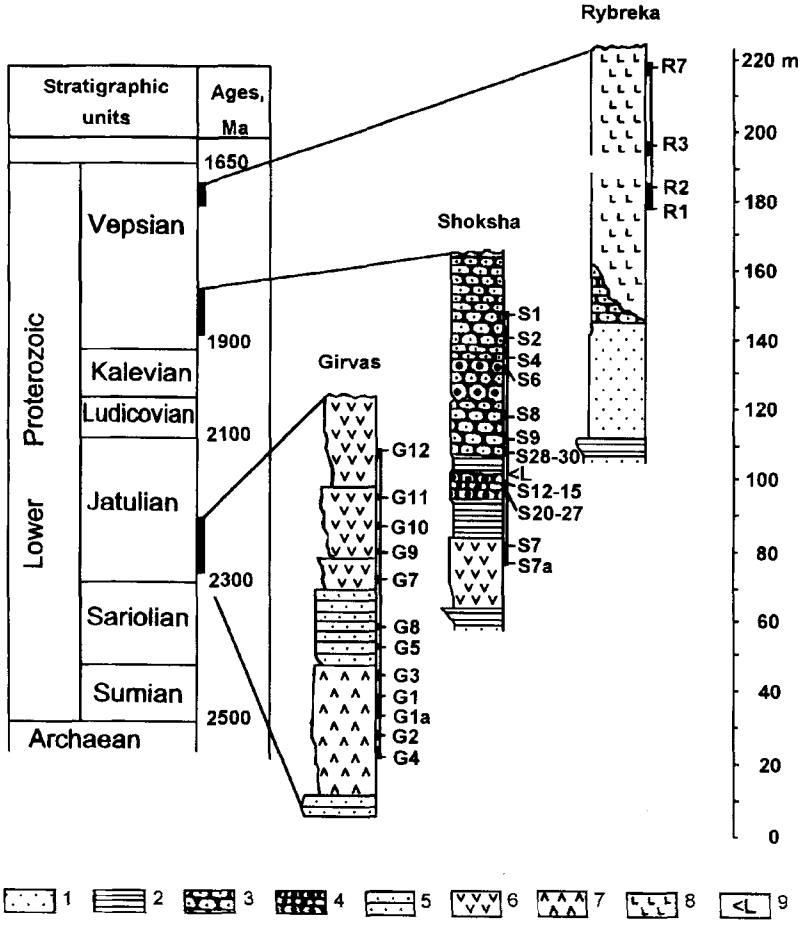


Figure 2. Stratigraphic columns of the outcrops investigated and their position in the stratigraphic scheme of Karelian Proterozoic (according to Semikhatov *et al.* 1991). 1: Petrozavodskian sandstones; 2: siltstones; 3: Shoksha quartzites; 4: conglomerates; 5: Jatulian quartzites; 6: basalts; 7: andesitic basalt; 8: gabbro-dolerites; 9: level of the intrusion of the gabbro-dolerite in the vicinity; 10: stratigraphic positions of the sampling sites.

the material sampled were carried out in the various laboratories participating, using spinner (JR-4) as well as astatic (LAM-24) magnetometers, both of Geofizika, Brno, production. The demagnetization equipment used was identical in all laboratories. Identical devices and technology have lead to minor scatter of the NRM-component directions isolated by cleaning (see Tables 1–3).

Due to the multicomponent behaviour of almost all specimens in the pilot study, eight to 15 demagnetization steps were used for the thermal and AF analysis of interest for all of the remaining specimens. A multicomponent analysis was carried out using Zijderveld diagrams (Zijderveld 1967) and stereoplots (Figs 3–12). A least-squares method was used for calculating NRM-component directions (Kirschvink 1980). The directions, based on three or more points on the diagram, and calculated with an error of 10° or less have been accepted. The polar equi-areal projection was used to plot a distribution of characteristic remanent magnetization (ChRM) directions (Figs 13 and 14). The mean directions and Fisher statistics were calculated at the sample level.

All ChRM-directional data and rock-magnetic results were collected and analysed in the palaeomagnetic laboratories of VNIGRI, St Petersburg and the Institute of Physics of the Earth, Moscow.

4 MAGNETIC CLEANING AND ROCK-MAGNETIC RESULTS

4.1 Rybreka gabbro-dolerite sill

Thermal and AF cleaning of specimens from the Rybreka sill reveal four components of magnetization. For the first group of specimens taken from the upper part of the sill, a characteristic magnetization (labelled component A_N) was isolated at temperatures above 500 °C or in alternating fields exceeding 40 mT (Fig. 3). The directions of the low-temperature (sometimes unstable) component are different from the direction of the recent field. This eastward component is labelled component C_N. Its mean direction is $D = 80.2°$, $I = 52.4°$ ($k = 26.3$, $\alpha_{95} = 13.7°$).

Generally, there is very good consistency between directions obtained during AF or thermal demagnetization (see Figs 3a,b). Thermal demagnetization, however, results in better definitions of the NRM components (Table 1). These results, together with other ones, have been included in the calculation of the component A_N.

The mean direction of component A_N for all sill samples is $D = 2.7°$, $I = 18.1°$ ($k = 51.3$, $\alpha_{95} = 7.4°$, seven samples, 12 specimens); see Table 2, first entry.

Seven samples taken from the lower part of the outcrops (i.e. from the central part of the sill; the lower part of the sill is not exposed here) yield a mean direction of $D = 221.6°$, $I = 14.6°$ ($k = 128.8$, $\alpha_{95} = 4.7°$); see Table 2, second entry. This magnetization component, labelled B_R, was isolated at temperatures above 400 °C for samples from site R2, and above 550 °C and in alternating fields exceeding 30 mT from site R1 (Fig. 4). At lower temperatures a magnetically soft component termed '*D*' with $D = 117.4°$, $I = -34.2°$ (not listed in Table 2) was isolated in two samples only (Fig. 4). In the remaining five samples, the low-blocking temperature–low-coercivity component identified coincides with the direction of the recent local geomagnetic field.

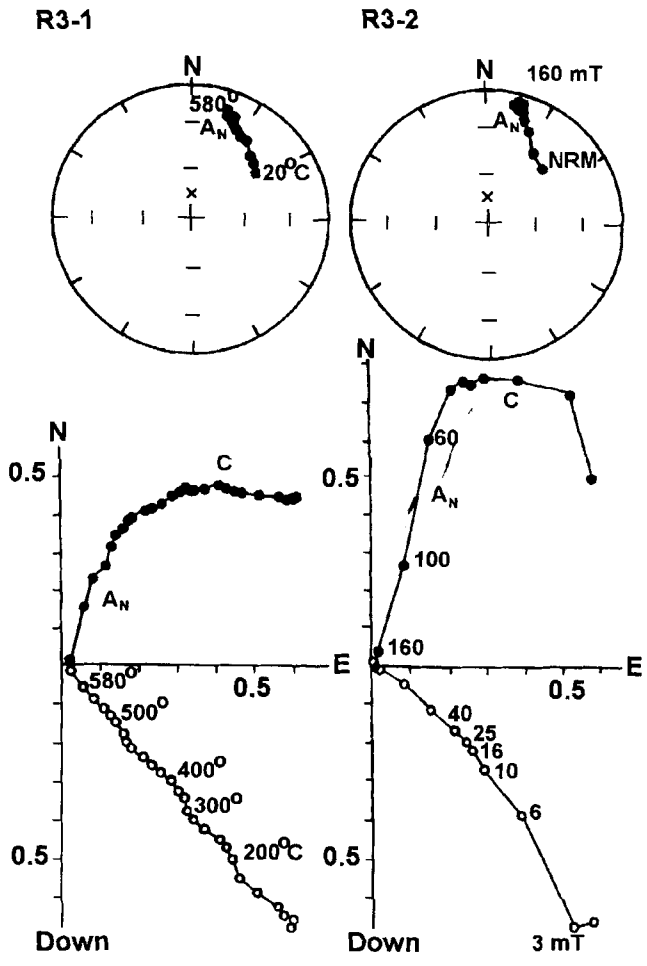


Figure 3. Equal-area stereographic projections and orthogonal projections (Zijderveld diagrams) of the NRM vector end-points for specimens of the Proterozoic rocks from South Karelia during thermal and AF demagnetization treatments. For all orthogonal projections, closed (open) symbols represent projection onto the horizontal (vertical) plane, respectively, where N is North, E is East, and Down is vertical down. Values are given in normalized units. Closed (open) symbols on stereographic projections represent projections on the lower (upper) hemispheres, respectively. NRM-component indices (A, B, C, D, E, F and G) isolated during demagnetization are plotted behind the intervals of their isolation. Values of the alternating field (mT) and temperatures (°C) are also shown. The figure shows Rybreka gabbro-dolerites, site R3, specimens R3-1 and R3-2.

Magnetic properties of rocks from the sampled part of Rybreka sill do not differ significantly for the above-mentioned two groups of samples. The magnetic susceptibility of the specimens varies from 7.5×10^{-2} to 11×10^{-2} SI units. Intensities of the NRM vary from 2.3 to 4.1 A m^{-1}, the mean value of the Koenigsberger's ratio $Q_n = 1.4$, and the ratio NRM/TRM ($= Q_{rt}$) varies from 0.1 to 0.7 (TRM was acquired in a field of 0.5×10^{-4} T). Of course, seven is too small a number of sites to trace the variations of the rock-magnetic properties across the sill.

Thermomagnetic analysis, $J_s(T)$ and $J_{rs}(T)$, of the rocks and investigation of the ferromagnetic mineral fractions reveals magnetite as the sole carrier of magnetization. No new mineral phases were formed during heating, thus indicating the presence of stable magnetite as carrier of the ChRM (Fig. 5c).

Table 1. Comparison of some selected AF and thermal demagnetization results defining component A_N for South Karelian Proterozoic rocks.

Rock types and locality	Site indices	Number of specimens	Isolation interval for component A_N	D (°)	I (°)	K	α_{95} (°)
Gabbro-dolerite	R3	5	500–590 °C	4.1	19.8	64.2	7.8
Rybreka	R3	5	>60 mT	3.3	17.7	22.6	13.2
Basalt, Shoksha	S7	3	500–590 °C	336.5	37.2	14.6	21.2
	S7	3	>75 mT	338.2	24.7	13.9	21.6
Basalt, Girvas	G3	4	150–570 °C	3.5	21.6	27.8	13.3
	G3	4	5–100 mT	3.3	22.0	21.1	15.2

Note. D, I: declination and inclination of the ChRM mean direction; K: precision parameter for mean direction; α_{95}: confidence circle radius for 95 per cent probability level.

Table 2. Palaeomagnetic directions from Vepsian rocks, Rybreka and Shoksha localities, South Karelia (61.5°N, 35.5°E).

Rock unit, NRM component	Isolation interval		Site/core/ specimen	Geographic coordinates		Stratgraphic coordinates		K	α_{95} (°)
	thermal (°C)	AF (mT)		D (°)	I (°)	D (°)	I (°)		
1. Rybreka sill, A_N	>500	>60	2/7/12	1.9	14.5	2.7	18.1	51.3	7.4*
2. Rybreka sill, B_R	400–600	>100	2/7/9	222.5	15.7	221.6	14.6	128.8	4.7*
3. Rybreka sill, C_N	100–400	10–40	1/4/4	75.5	54.0	80.2	52.4	26.3	13.7
4. Shoksha quartzite, A_N	300–570	20–100	7/16/18	4.2	22.1	6.3	31.2	36.4	5.8*
5. Shoksha quartzite, A_R	550–670	>40	2/5/6	163.5	−26.0	161.3	−32.7	53.7	8.5*
6. Shoksha quartzite, $A_N + A_R$			9/21/24	1.4	23.9	1.0	31.9	27.4	5.9*
7. Shoksha quartzite, B_R	100–600		2/3/6	193.4	40.0	192.6	31.9	37.4	13.2*
8. Shoksha quartzite, C_N	570–670		4/7/9	80.4	40.8	85.6	41.7	16.1	13.2
9. Shoksha quartzite, C_R	570–670		4/6/7	294.2	−41.9	299.4	−38.5	63.9	7.6
10. Shoksha basalt, C_N	300–440		1/3/4	103.1	41.6	109.6	39.5	39.5	12.4
11. Shoksha, all rocks, $C_N + C_R$			9/16/20	94.5	42.5	102.5	41.2	17.2	8.3*
12. Shoksha quartzite, D	400–670		3/4/6	71.8	−37.3	66.6	−34.1	73.0	8.2
13. Shoksha basalt, A_N	550–650	>75	2/7/9	345.1	22.1	344.1	29.5	23.9	10.8*

Notes. Sample (core) statistical level used. * used for pole calculation, see Table 4; for D, I, K, α_{95}, see note for Table 1.

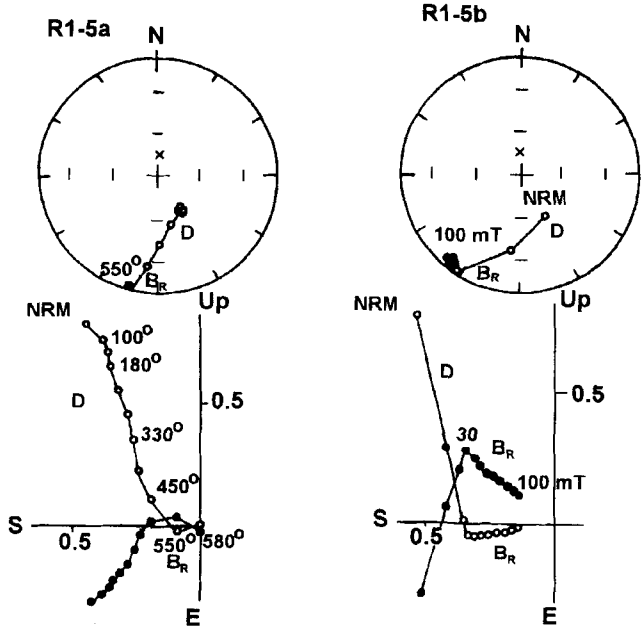


Figure 4. As Fig. 3, for Rybreka gabbro-dolerites, site R1, sample R1-5, specimens R1-5a and R1-5b.

4.2 Shoksha quartzites

NRM values of the Shoksha quartzites vary from 1.0 to 12.7 mA m^{-1}; the mean value of Q_n is 2.0.

Thermal demagnetization of red quartzites from site S2 was carried out in nine steps (T = 100, 200, 260, 320, 360, 440, 500, 550 and 600 °C). Nine specimens from four samples (S2-7 to S2-10) were investigated.

Fig. 6 illustrates the multicomponent behaviour of the magnetization, which shows at least three different NRM components. The first characteristic directions (samples S2-7, S2-8 and S2-10) are grouped in the northern quadrant of the stereoplot with shallow inclinations (component A_N, Fig. 6c). A second southwesterly shallow downward-directed remanence (component B_R) was isolated in the temperature range of 200° to 600 °C in specimens from sample S2-9, and for one specimen from sample S2-7 (Fig. 6b). Above 600 °C a third component of dual polarity (C_N, C_R) with NW–SE direction declination and moderately steep inclination can be observed; however, its true direction can be obtained only by complete demagnetization. These three components identified in the samples from site S2 are similar to the components isolated in the Rybreka sill. Shoksha quartzites sampled from other sites, S1, S4, S6, S8 and S9 (see examples on Fig. 7), appear to exhibit behaviour during thermal demagnetization similar to that of site S2.

For more precise isolation of the high-temperature NRM

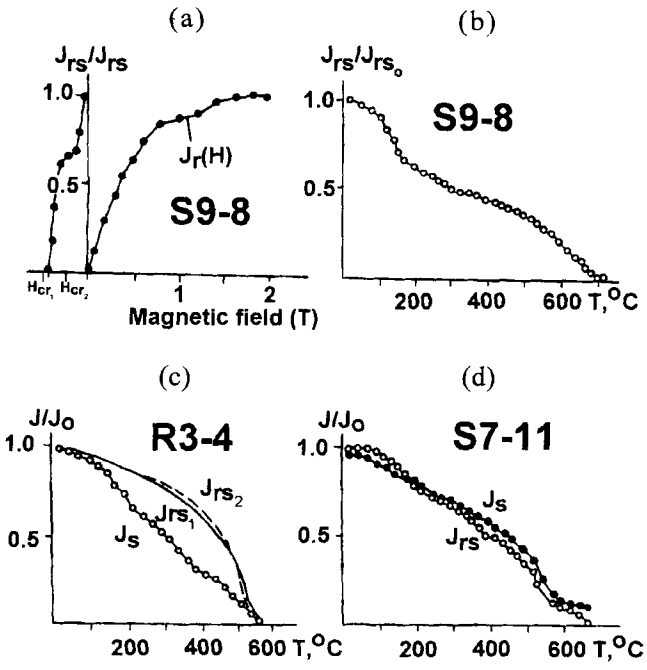


Figure 5. Examples of the results of the isothermal magnetization (a) and thermal demagnetization (b, c and d) experiments for Vepsian rocks, S. Karelia. Isothermal saturation magnetization J_s and isothermal saturation remanent magnetization J_{rs} curves are plotted for specimens from Rybreka sill (R3-4), Shoksha quartzites (S9-8) and Shoksha basalt (S7-11). J_{rs1}: first heating; J_{rs2}: second heating. Curves without data points have been drawn automatically during continuous thermal demagnetization.

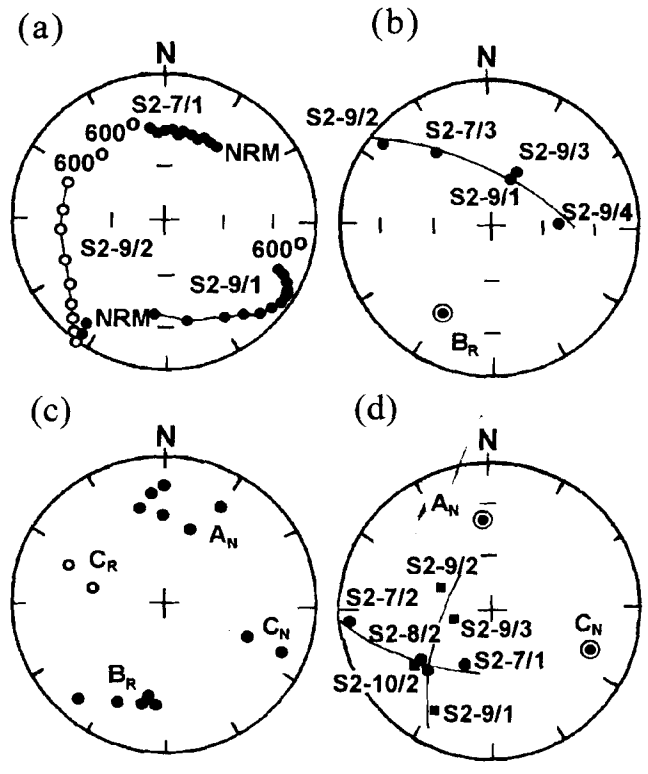


Figure 6. Shoksha quartzite, site S2: four specimens from sample S2-9 (components B_R and C_N, C_R), and specimens from samples S2-7, S2-8 and S2-10 (component A_N). (a) examples of the direction changes during thermodemagnetization; (b), (d) distribution of normals to demagnetization circles for B (200–500 °C), A (200–570 °C) and C (> 570 °C) components. Mean directions of the A, B and C components (c) were calculated using Halls' method (Halls 1978).

components of the Shoksha quartzites, very detailed thermal demagnetization experiments on specimens from sites S12-15 and S20-30 have been carried out. The example plotted in Fig. 7 demonstrates that haematite is the single carrier of the C-component and partly of the A-component. Unblocking temperatures of the B-components, and in some cases of the A-component, suggest that magnetite is the carrier. However, petrographic investigations establish only very rare occasional magnetite grains among accessories, whereas these rocks contain 0.5–1.0 per cent Fe_2O_3 as coatings on the detrital quartz grains, microcrystallites inside these grains and needle and tabular (~0.01 mm size) grains associated with the regenerated quartz crystals (Simanovich 1966). Perhaps these three haematite generations are responsible for the A, B and C-components of the Shoksha quartzites. Multicomponent analysis of all samples of the Shoksha quartzites permits calculation of the mean directions of these components and also of a fourth component A_R with a southeast upward-direction (Table 2).

The temperature decay of saturation magnetization confirms haematite to be the principal magnetic carrier for the Shoksha rocks (Figs 5a,b,d). Thermomagnetic curves $J_{rs}(T)$ show one or two kinks in the interval from 100° to 300 °C and a Curie point near 675 °C (Fig. 5b). The existence of two components of the magnetic carrier is clearly visible in the $J_r(H)$ curves and the $J_{rs}(H)$ curve (Fig. 5a). These components have $H_c = 275$ and 425 mT respectively. The petrographic data mentioned above permit the association of these components with two different forms of haematite.

4.3 Shoksha basalts

Specimens sampled from basaltic lava flow are characterized by NRM, ranging from 0.15 to 0.25 A m^{-1}; the mean Koenigsberger's ratio is $Q_n = 1.49$.

The result of detailed AF and thermal demagnetization experiments on basaltic lava specimens from sample S7-12 is shown in Fig. 8. Two, perhaps three, components may be observed in the Zijderveld plots. The ChRM is isolated above 400°–550 °C and above 40–75 mT. The ChRM directions for site S7 obtained by thermal and AF cleaning are given in Table 1. The thermal demagnetization data indicate that magnetite is the main carrier of NRM, besides a subordinate haematite contribution (Fig. 5d). Rocks from the other site, S7a, yield additionally a remanence component C, directed to the east downwards. This component was isolated at 300°–440 °C, and is similar to that found in the Shoksha quartzites.

In summary, it can be stated that the basalts underlying the Shoksha quartzite show two magnetization components, the ChRM being determined better by thermal demagnetization. The first, high-temperature, A component yields the mean values $D = 344.1°$, $I = 29.5°$, $k = 23.9$, $\alpha_{95} = 10.8°$, and the mean direction of the second, moderate-temperature, C-component is $D = 109.6°$, $I = 39.5°$ (see Table 2).

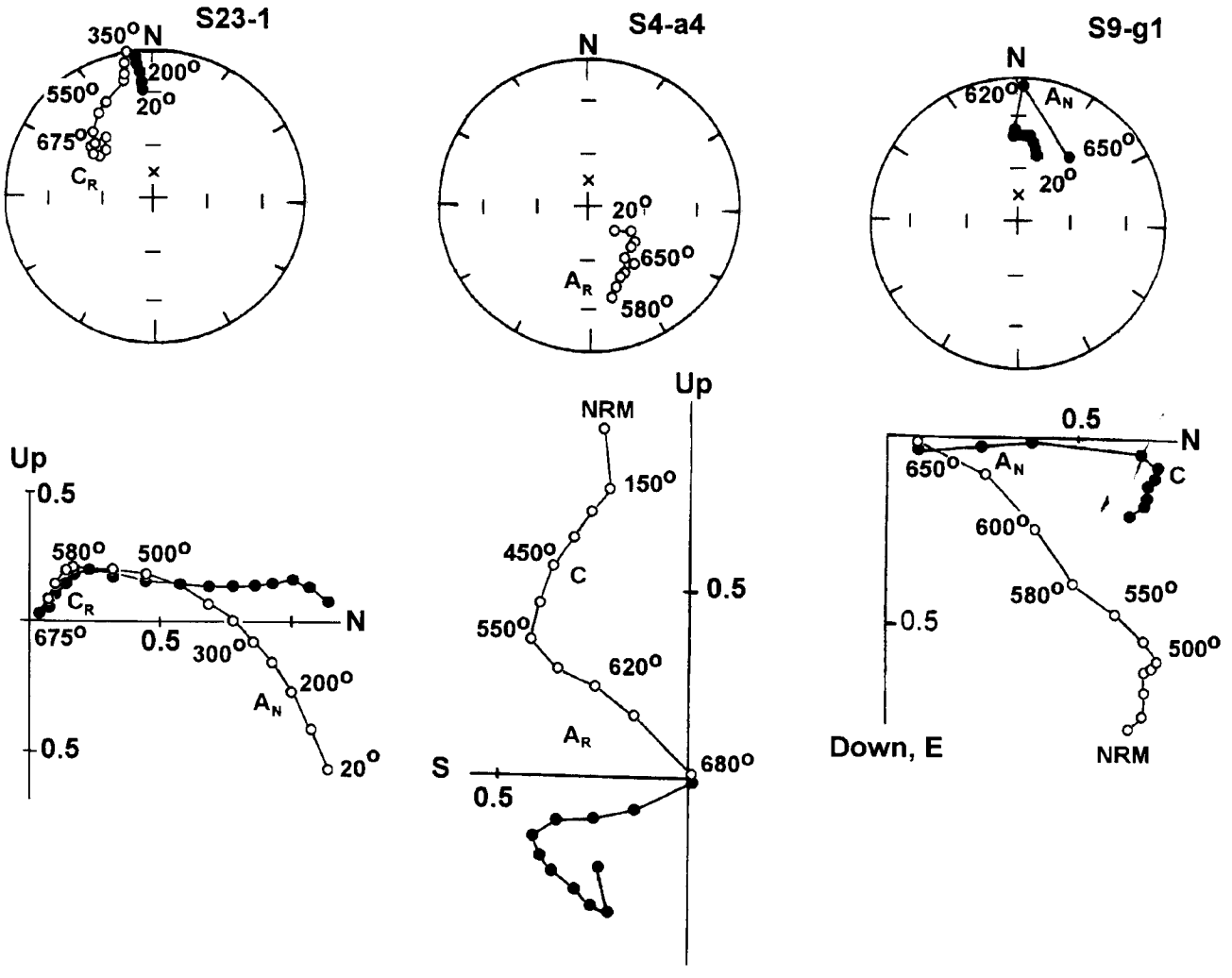


Figure 7. As Fig. 3, for Shoksha quartzites, sites S4, S9 and S23-1.

4.4 Girvas basalts

Based on the magnetic properties and behaviour during the demagnetization, the Girvas basalts can be grouped into two. The first group is characterized by relatively high values of the NRM (2.3 to 4.8 A m^{-1}) and magnetic susceptibility (2.7 to 69.5 $\times$ 10^{-3} SI), the mean values of the Q ratio being $Q_n = 1.1$, $Q_{rt} = 0.52$, respectively. Almost all samples taken from Lower Jatulian andesitic basalt (G1, G1a, G2, G3 and G4) fulfil the above characteristics.

The majority of the specimens investigated are magnetically very soft: AF-demagnetization peak fields of 10 mT remove up to 90 per cent of the intensity of the initial NRM. However, a harder NRM component, displaying more stable directions, is defined at higher fields. Nevertheless, the presence of a large soft magnetic phase does not facilitate the estimation of the ChRM directions by means of AF cleaning. Thermal cleaning, however, yields better results (Fig. 9). Heating experiments suggest that the main ferromagnetic phases are magnetite and small quantities of haematite (Fig. 10). The shape of the $J_s(T)$ and $J_{rs}(T)$ curves during the initial (J_{s1}) and second (J_{s2}) heatings are rather similar. Nevertheless, the specimen is magnetically unstable ($J_{s2}/J_{s1} = 1.2$). The formation of new magnetite has evidently occurred due to transformation of the paramagnetic component. There are also some peculiarities in the $J_s(T)$ and $J_{rs}(T)$ curves, such as bending between 100° and 300 °C, pointing to the presence of a low-temperature magnetic carrier (Fig. 10, specimen G1-14). The similarity of the $J_{s1}(T)$ and $J_{s2}(T)$ curves means that this low-temperature magnetic carrier is thermally stable, or regenerates during the cooling. Possibly this peculiarity reflects titanomagnetite solid solution regeneration.

All samples taken from the Middle Jatulian flows from sites G7, G9, G10 and G11 belong to the second group of rocks. These rocks have low values of NRM intensity (0.003 to 0.13 A m^{-1}) and low magnetic susceptibility (20 to 40 $\times$ 10^{-5} SI), with wide variations of Q_n (1.0 to 12). Thermal decay curves $J_s(T)$ suggest that the magnetic carrier is haematite (Fig. 10). Macroscopically, all these rocks look strongly oxidized; the presence of haematite flakes and pigment has been established by microscopic analysis. Thermal cleaning reveals two components of the NRM, haematite being the main carrier (Fig. 11).

Haematized quartzites, which cover Lower Jatulian flows, have magnetic properties rather similar to those of the Middle Jatulian oxidized basalts: the NRM intensity varies from 2.9 to 30 mA m^{-1} and values of magnetic susceptibility are 0.08 to 0.5 $\times$ 10^{-3} SI units. Thermal decay curves of $J_s(T)$ (Fig. 10)

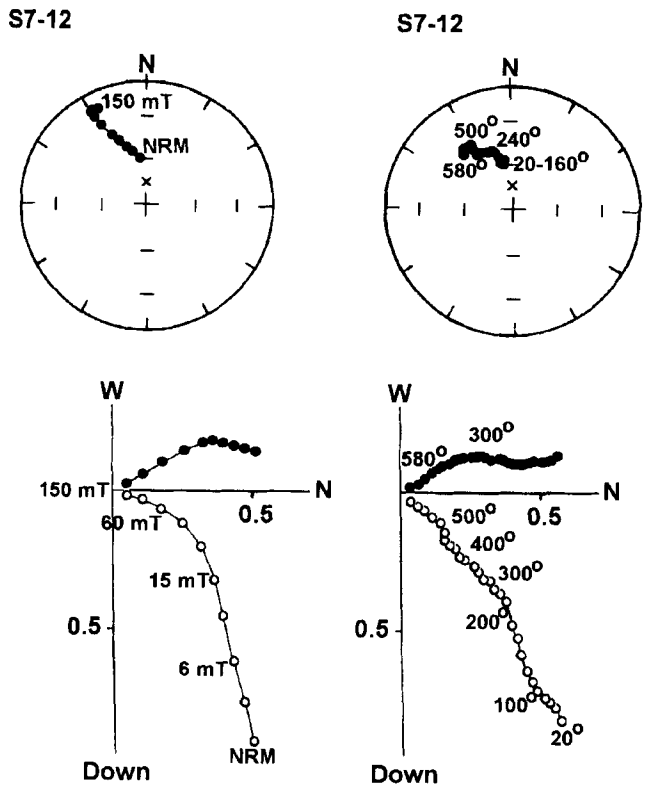


Figure 8. As Fig. 3, for Shoksha basalt, site S7.

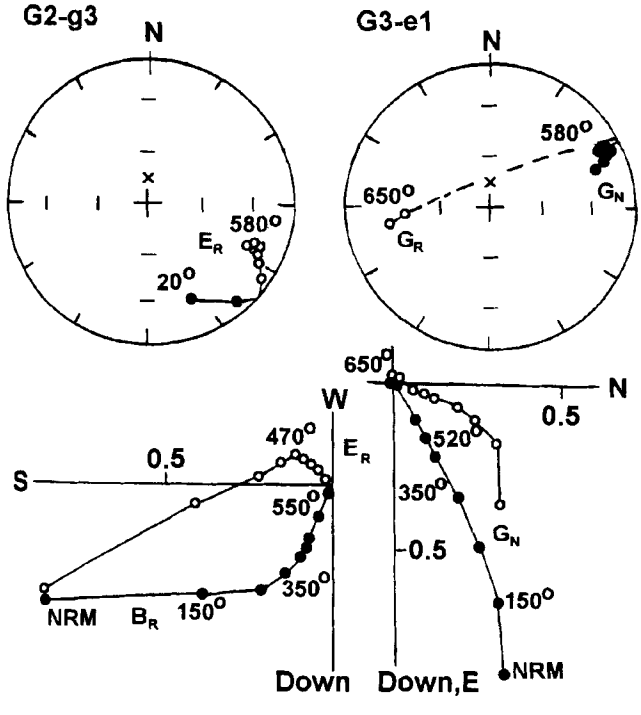


Figure 9. As Fig. 3, for Lower Jatulian Girvas andesitic basalt, sites G2 and G3.

and Zijderveld plots (Fig. 12) suggest haematite as the magnetic carrier of rocks ($T_c = 675\,°C$).

One specimen from sample G5-15 was subjected to detailed thermal cleaning. The Zijderveld plot in Fig. 12 shows three components of NRM. Although this Zijderveld diagram

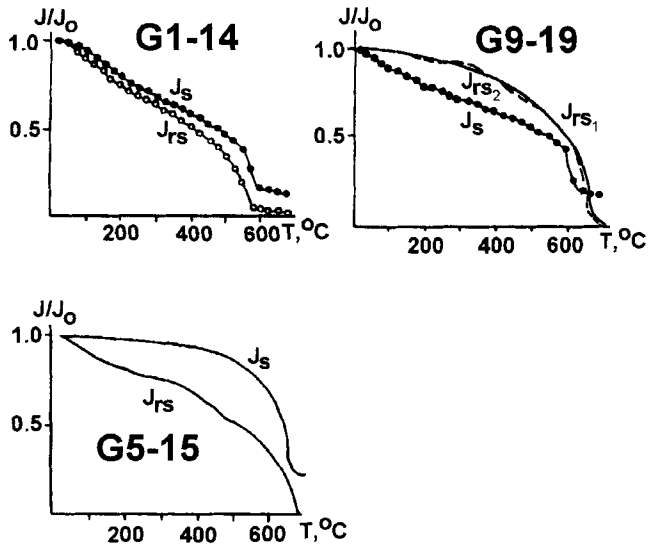


Figure 10. Examples of the results of the thermal demagnetization experiments for Lower Jatulian basalt (G1-14), Middle Jatulian basalt (G9-19) and quartzite (G5-15).

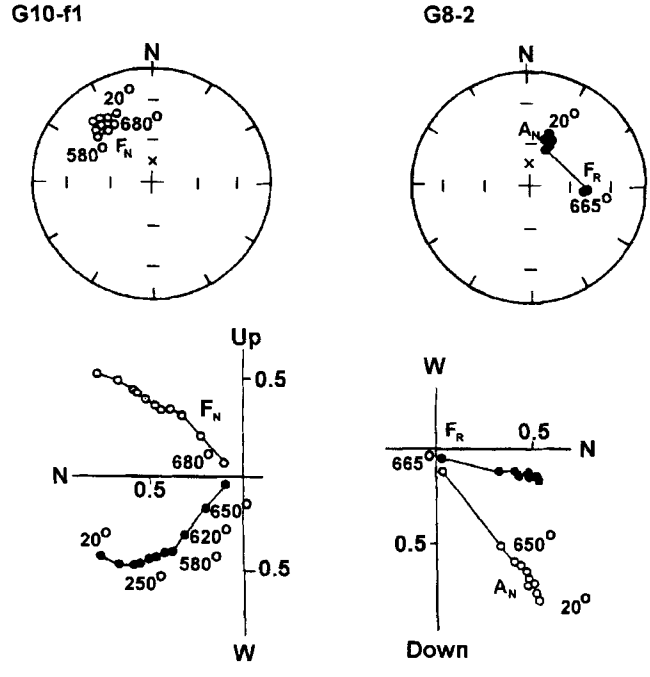


Figure 11. As Fig. 3, for Middle Jatulian basalts, sites G8 and G10.

demonstrates an overlap in the blocking-temperature spectra, it can be seen that the first unstable component is destroyed at temperatures of about 200 °C, the second has blocking temperatures in the range 200°–520 °C and the third, with a steep inclination and eastward declination, is completely removed at 680 °C. The distributions of the component directions for specimens G5-15 and G5-16 are shown on the stereonet of Fig. 12, which demonstrates that these directions lie along the great circle connecting the directions of the components A_N and F_R, typical for the Middle Jatulian basalts (for example, G8-2, Fig. 11, and G7-2, Fig. 12). That is, only two components, A_N and F_R, with broad overlapping unblocking-temperature spectra, really exist in these specimens.

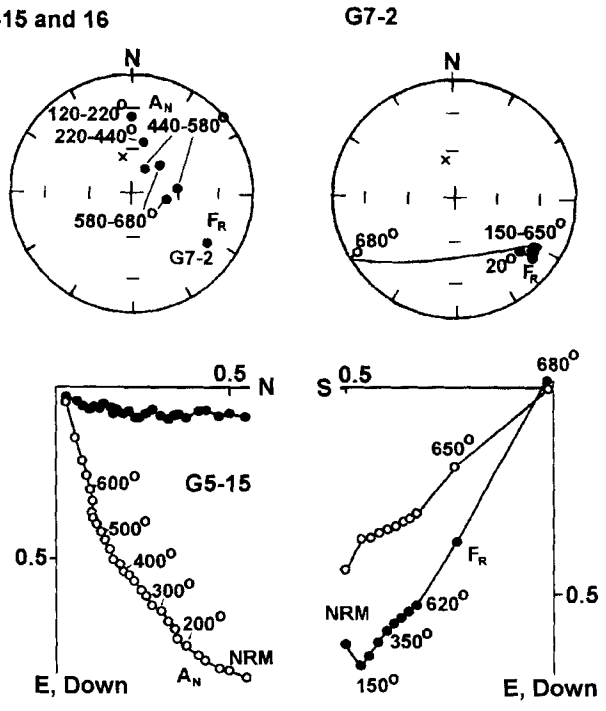


Figure 12. As Fig. 3, for Middle Jatulian quartzite (specimens from site G5) and basalt (site G7).

In summary, the Girvas basalts and interbedded quartzites carry four components: the single-polarity component A_N and the dual-polarity components E, F and G. Their mean directions, statistics and isolation intervals are shown in Table 3.

5 PALAEOMAGNETIC ANALYSIS

In most cases, the rocks investigated show multicomponent magnetic behaviour. These NRM components differ with respect to unblocking temperature and coercivity spectrum (Tables 2 and 3).

The main problem is to assign ages to these components. Unfortunately, it was not possible to apply any fold test, due to minor tilting of the strata. Nevertheless, all NRM-component directions are also represented in Tables 1–3 in their stratigraphic coordinates, although some of these components may be post-folding. Furthermore, the close temporal proximity of deposition of the sediments and of emplacement of the magmatic bodies means that the results of the contact test are inconclusive as well. However, there is a possibility of estimating the ages of some of the NRM components.

5.1 Rybreka sill and Shoksha rocks

Component A dominates in specimens from the Rybreka sill (Figs 3 and 13), and in specimens from the basaltic lavas underlying the Shoksha quartzite (Table 2, entries 1 and 13). Red quartzites and quartzitic sandstones overlying these lavas and interrupted by the Rybreka sill also carry component A (Figs 7 and 13, Table 2, entry 4). Its age is constrained by the ages of the lavas and of the sill. In the Shoksha quartzites, this component has dual polarity (Figs 7 and 13). Statistics of the normal and reverse polarities of these rocks correspond to class I, according to the classification of the reversal test of McFadden & McElhinny (1990). On the other hand, this test appears more determinable (class C), if we use the A_N component for the lavas: here the statistics correspond to the A_N component for lavas and the A_R component for quartzites. As to the A_N component for quartzites, we suppose it is later in origin.

Component A_N is the high-temperature component in the outer, fine-grained part of the sill, when the stable, fresh-looking stochiometric ($T_c = 579\,^\circ C$) magnetite is the carrier of this component. This particular feature and relatively high values of the Q_n and Q_{rt} ratios (see 4.1) point to the high probability of a thermoremanent origin of this NRM component. Hence, it is possible to say that the age of the A_N component is the same as the age of the sill (1770 ± 12 Ma). The age of the presumably primary A_R component identified in the Shoksha quartzites and A_N in basalts is therefore ~ 1800 Ma, according to the K–Ar age of the underlying Petrozavodskian sandstones.

Component B_R is isolated as a high-temperature component in rocks from the central part of the Rybreka sill. There are a large number of monzonite veins related to the last stage of

Table 3. Palaeomagnetic directions from Jatulian rocks, Girvas, South Karelia, (62.5°N, 34.0°E).

Rock unit, NRM component	Isolation interval		Site/core/ specimen	Geographic coordinates		Stratgraphic coordinates		K	α_{95} (°)
	thermal (°C)	AF (mT)		D (°)	I (°)	D (°)	I (°)		
14. Lower and Middle Jatulian lavas, A_N	150–570	10–100	7/15/21	1.8	45.2	3.0	29.3	20.6	8.0*
15. Lower Jatulian quartzite, A_N	550–670	>100	2/5/5	357.0	50.1	355.5	38.0	20.0	14.0
16. Lower Jatulian lavas, E_R	350–650		2/3/8	111.3	−50.9	127.1	−43.5	9.3	26.5
17. Lower Jatulian lavas, E_N	360–450		1/3/3	286.0	45.1	297.9	39.9	11.0	23.9
18. Middle Jatulian lavas, F_N	250–600	10–80	3/5/11	332.2	−42.1	331.2	−47.5	26.0	12.3
19. Middle Jatulian lavas, F_R	>520		2/5/8	146.5	24.1	143.0	36.6	20.0	14.0
20. Middle Jatulian lavas, $F_N + F_R$			5/10/19	329.7	−33.2	326.8	−42.1	22.0	9.5*
21. Lower Jatulian lava, G_N	150–450		3/5/8	70.8	50.2	65.9	45.3	14.8	16.3
22. Lower Jatulian quartzites, G_N	625–675		2/3/5	91.5	60.2	72.7	61.3	27.1	15.5
23. Lower Jatulian lava, G_R	>600	>100	3/3/4	250.2	−44.4	254.7	−45.1	7.7	29.1
24. Lower Jatulian lava and quartzite, $G_N + G_R$			8/11/17	82.8	52.0	69.9	49.8	14.4	11.1*

Notes. Sample (core) statistical level used. * used for pole calculations, see note for Table 4; for D, I, K, α_{95}, see Table 1.

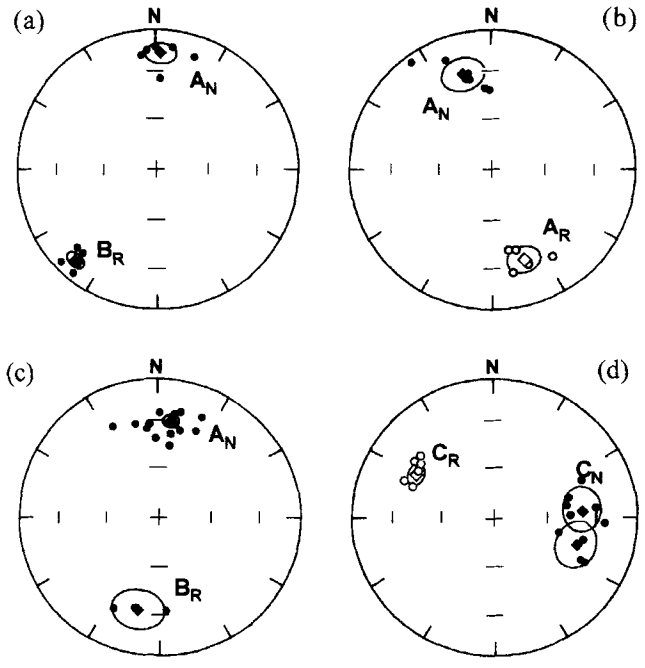


Figure 13. ChRM directions for Vepsian rocks, S. Karelia. Circles of confidence are shown around the mean directions. (a) Rybreka sill, components A_N and B_R; (b) Shoksha basalt, component A_N, and Shoksha quartzites, component A_R; (c) Shoksha quartzites, components A_N and B_R; (d) Shoksha quartzites and basalt, components C_N and C_R.

sill emplacement (Bibikova *et al.* 1990). Similar veins are observed also in the Shoksha quartzites. Hence, component B_R, both in the sill and in the quartzites, may be considered slightly younger than component *A*. The dual-polarity component *C* isolated in the Shoksha quartzites at high temperatures is assumed to be secondary, although there is no clear evidence for its secondary origin. Furthermore, after reversing the polarity of this component, its pole falls near the APWP track for 2120–2450 Ma. However this would indicate a Jatulian age for the Shoksha quartzite, which is not compatible with geological evidence. The last indirect argument for secondary origin of component *C* is that identified in the Rybreka sill also, being a soft ($H_c = 6$–41 mT) and low-temperature ($T_{ub} = 100°$–$350\,°C$) component (Fig. 3).

5.2 Jatulian lavas and quartzites

Lower Jatulian lavas carry two high-temperature (>600 °C) components, G_R and $E(E_N, E_R)$. Component *E* is defined only poorly (Table 3) and is not discussed any further. As for component G_R, its direction is opposite to that of the moderate-temperature component G_N in this lava (Fig. 9). The same high-temperature component has been isolated in the interbedded quartzites (Table 3, entries 21, 22, 23). The possible primary origin of components G_N and G_R is not supported by the reversal test (class I—see Table 3, entries 21, 22, 23); however it is favoured by a positive consistency test (three types of rocks) and high values of Q_n and Q_{rt} for rocks (see 4.4).

Middle Jatulian lavas are characterized by a high degree of high-temperature oxidation of the outer parts of bodies sampled. Haematite occurring in these parts and in the interbedded quartzites carries the high-temperature component F_R and moderate-temperature component F_N (Figs 11, 12 and 14, Table 3 entries 12, 18 and 19). In some cases, the direction of the moderate-temperature component in the quartzites lies along a great circle connecting directions of the components F_N and A_N (Fig. 12). The NRM-component A_N has the same direction as component A_N for Vepsian rocks and occurs in many Lower and Middle Jatulian rocks as a magnetite component A_N which overprints (Table 3, entries 14 and 15) the older dual-polarity components *F* and *G*. Because component *F* is associated with Middle Jatulian rocks and component *G* with Lower Jatulian, it is suggested that component *G* is older than component *F*. Component A_N may be assigned to the Svecofennian tectonomagmatic event.

Consequently, these four components can be correlated with the following time intervals:

component *A*: 1770–1850 Ma, Svecofennian; $D = 341(161)$–$6°$, $I = 18(-18)$–$38°$;
component *B*: 1500–1650 Ma, Post-Svecofennian; $D = 193$–$222°$, $I = 15$–$32°$;
component *F*: 2200 Ma, Middle Jatulian; $D = 323(143)$–$331°$, $I = -37(37)$– $-48°$;
component *G*: 2300–2400 Ma, Early Jatulian; $D = 66(246)$–$75°$, $I = 45(-45)$–$61°$.

Table 2 shows that quartzites and magmatic rocks at Shoksha carry other components, *C* and *D* (entries 3, 8, 9, 11 and 12), with a rather wide spectrum of unblocking temperatures. Their origin and age are unclear. The statistically significant palaeomagnetic poles are listed in Table 4.

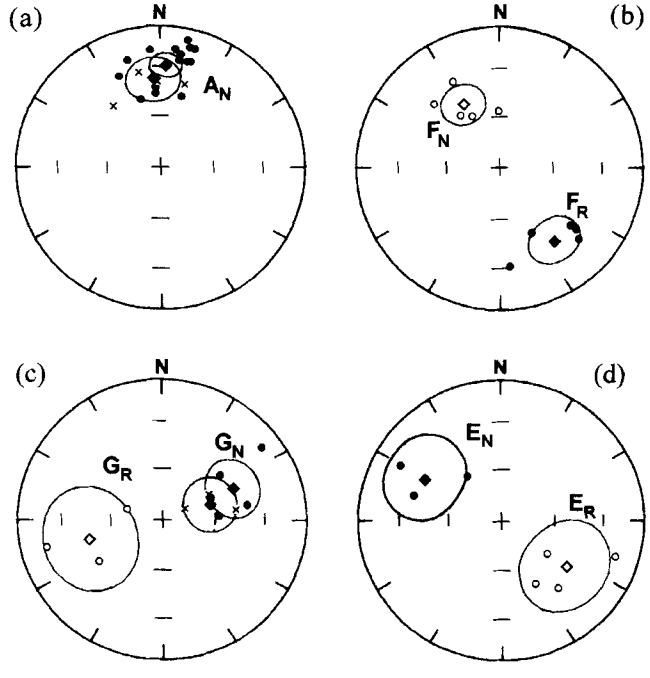


Figure 14. ChRM directions for Jatulian rocks, S. Karelia. Directions for sedimentary rocks are denoted by crosses. (a) Middle Jatulian Girvas basalts (dots) and quartzites (crosses), component A_N; (b) Middle Jatulian Girvas basalts, components F_N and F_R; (c) Lower Jatulian Girvas basalts (dots) and quartzites (crosses), components G_N and G_R; (d) Lower Jatulian Girvas basalts, components E_N and E_R.

Table 4. Summary of the palaeomagnetic poles, South Karelia.

Nr.	Rock unit, geological age	Assigned magnetic age (Ma)	NRM component	Polarity	Pole position				
					Φ (°N)	Λ (°E)	dm (°)	dp (°)	A_{95} (°)
1.	Rybreka sill, Vepsian	1770	A_N	7N	37.8	210.6	7.7	4.0	5.5
2.	Rybreka sill, Vepsian	1580*	B_R	7R	13.9	171.3	4.8	2.4	3.2
4.	Shoksha quartzite, Vepsian	1770	A_N	16N	45.1	206.4	6.5	3.6	4.8
5.	Shoksha quartzite, Vepsian	1850	A_R	5R	44.2	241.5	9.6	6.1	7.6
7.	Shoksha quartzite, Vepsian	1580*	B_R	3R	10.5	201.8	14.9	8.4	11.2
11.	Shoksha rocks, Vepsian	?	C_N, C_R	10N, 6R	14.9	102.8	10.1	6.2	8.0
13.	Shoksha basalt, Vepsian	1850	A_N	7N	42.9	235.1	12.3	6.9	8.5
14.	Girvas lavas, Jatulian	1800	A_N	15N	43.1	211.5	8.9	4.9	6.6
20.	Girvas lava and quartzite, Middle Jatulian	2200	F_N, F_R	5N, 5R	−0.8	245.4	11.6	7.2	9.1
24.	Girvas rocks, Lower Jatulian	2400	G_N, G_R	3N, 8R	−36.0	307.4	14.9	9.9	12.0

Notes. * these ages are estimated from the APW paths for Precambrian (Fig. 14); for the others, see text. Φ, Λ: latitude and longitude of palaeomagnetic pole position; dm, dp: semiaxes of the confidence oval around the pole; A_{95}: circle of confidence for poles.

6 DISCUSSION OF THE RESULTS AND CONCLUSIONS

The main result of this investigation, and that most interesting for Precambrian palaeomagnetism, is the set of new palaeomagnetic pole positions for that time interval.

The new pole positions are in good agreement with the 2120–1750 track of the APWP of the Fennoscandian Shield constructed by Elming *et al.* (1993) (Fig. 15) on the basis of the revised palaeomagnetic database for this region (Elming *et al.* 1995).

New palaeomagnetic results for the Lower Proterozoic rocks from the Fennoscandian Shield obtained after the above-mentioned compilations (Mertanen 1995; Torsvik & Meert 1995; Krasnova & Gooskova 1995; Fedotova *et al.* in preparation) and listed in Table 5 (poles 25–29, 35) are in good agreement with the 2120–1750 Ma track also (Fig. 15). However, the situation is quite different for the pre-Svecofennian part of the APWP: there is a clear deviation from this APWP observed for the Jatulian time interval (poles 20 and 24 and coeval poles from other authors, Table 5, poles 30–34, 37, 38). Perhaps this is a consequence of the previous lack of the 'grand mean poles' for the 2450–2120 Ma interval on Elming's APW path (Fig. 15). Thus the real APWP for this interval may be more complicated than the one drawn in Fig. 15. Alternatively, the Fennoscandian Shield might not

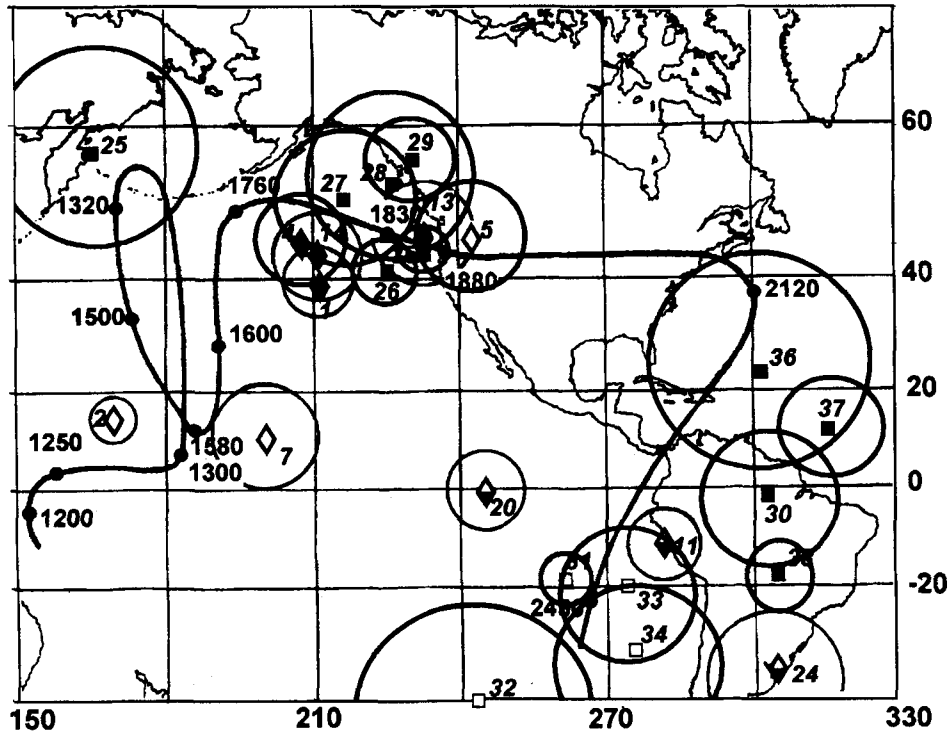


Figure 15. The APWP for Fennoscandia for the time interval 2.45–1.2 Ga (after Elming *et al.* 1993) and the new coeval palaeomagnetic poles. Black dots denote the key poles after Elming *et al.* (1993). New poles for the eastern Fennoscandia are denoted by diamonds (Table 4) and quadrangles (Table 5). Black figures: normal polarity; contour figures: reversed polarity; half-shaded figures: mixed polarity.

Table 5. Some new Early Proterozoic palaeomagnetic poles for eastern Fennoscandia.

Nr	Rock unit	Assigned magnetic age, Ma	Polarity	Pole position			Reference
				Φ (°N)	Λ (°E)	A_{95} (°)	
Karelian province (61.5°–66.2°N; 30.2°–36.2°E)							
25.	Tikshozero intrusion and dykes, comp. *B*	1700	17N	57.0	165.0	13.0	Krasnova & Gooskova (1995)
26.	Ropruchey sill	1770	10N	40.5	227.3	7.1	Fedotova *et al.* (in prep.)
27.	Lupzenga dykes, comp. *A*	1840	5N	51.4	216.2	9.4	Mertanen (1995)
28.	Suopera dykes, comp. *A*	1840	5N	51.1	230.6	6.1	Mertanen (1995)
29.	Tikshozero intrusion and dykes, comp. *A*	1900	13N	53.0	228.0	10.0	Krasnova & Gooskova (1995)
30.	Matozero dykes	2300	6N	−1.6	301.9	14.3	Fedotova *et al.* (in prep.)
31.	Burakovka gabbro-norites	2440	11R	−19.5	262.6	7.2	Fedotova *et al.* (in prep.)
32.	Tikshozero intrusion comp. *C*	2410	5R	−41.0	245.0	10.0	Krasnova & Gooskova (1995)
33.	Lupzenga dykes, comp. *D*	2440	13R	−20.4	276.1	13.5	Mertanen (1995)
34.	Suopera dykes, comp. *D*	2446	3R	−31.0	277.1	13.3	Mertanen (1995)
Belomorian and Kola provinces (67.2°–69.5°N; 29.5°–32.6°E)							
35.	Kolvitsa dykes	1900	9N	44.2	323.6	5.4	Fedotova *et al.* (in prep.)
36.	Kuetsyarvi Fm	2150	10N	24.7	300.8	17.6	Torsvik & Meert (1995)
37.	Voche-Lambina ultramafites	2200	14N	12.6	314.5	12.6	Fedotova *et al.* (in prep.)
38.	Main Range gabbro-dolerites	2430	10N	−16.7	304.8	8.3	Fedotova *et al.* (in prep.)

Note. Φ, Λ, A_{95}: see Table 4.

have been a single rigid body prior to 2120 Ma (Berthelsen & Marker 1986).

As for the poles 2, 7 and 11, which refer to the Shoksha and Rybreka NRM components *B* and *C*, they are located near the APWP: that is, near the 1300–1230 Ma old APWP track and near the Jatulian track. We stressed earlier that the age and polarity of pole 11 are very dubious. On the other hand, the presence of a strong *A* component in the Jatulian rocks (pole 14) and for pre-Jatulian rocks, investigated by other authors (Table 5, poles 25, 27, 28, 29, 35), indicates overprinting processes during the Svecofennian tectonomagmatic events, clearly seen over the whole eastern Fennoscandian shield. At the same time, two other poles, 2 and 7, both with reversed polarity, are in the vicinity of the 1580 Ma swath of the Post-Svecofennian loop (Fig. 15). It is possible that they correspond to some Post-Svecofennian (Jotnian) tectonomagmatic events; however, manifestations of these events are only known in the more westerly regions of South Karelia (Sokolov 1987). Assuming that the Fennoscandian Shield was stable from 2400 to 1800 Ma (Pesonen *et al.* 1989; Gaal & Gorbatschev 1987), the new data confirm a significant change in its position relative to the geographical coordinate system during this interval. Nevertheless, for further improvement of the APWP during the time interval 2400–1800 Ma, additional data are necessary. Furthermore, these data would be very important for resolving the problem of intracratonic horizontal movements, which still cannot be resolved by using the available palaeomagnetic data. According to Gorbatchev & Bogdanova (1993), rifting and break-up between 2500 and 2000 Ma strongly affected and partly dispersed the Archean Domain of the Fennoscandian Shield. The resultant basins were closed by collisional orogeny between 1950 and 1820 Ma, which possibly marked the formation of the Proterozoic Supercontinent (Piper 1987). The aim of the future palaeomagnetic investigation will be to check this model, to estimate the unknown width of these Proterozoic oceans, and to confirm the time of its cessation.

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